CS 498 QC ✧ Spring 2026

Homework 1

Due Tuesday, February 24th, 2026 at 9pm

- For this homework, you are encouraged to collaborate with your classmates. You should submit your solutions **individually**.
- **Submit your solutions electronically on Gradescope as PDF files.**
- Please format your solutions so that each problem begins on a new page and your name appears at the top of each page. While handwritten submissions are allowed, all homework submissions are **preferred to be typeset** using Latex or other equivalent programs.
- Problems marked ★ are optional and usually a little harder.

☞ Some important course policies ☜

- You are encouraged to collaborate with your classmates on homework problems. You are allowed to look at definitions of mathematical objects and general problem solving strategies as well. You should first make a serious attempt to solve the problems on your own. After that, you may use AI tools like ChatGPT, and you are encouraged to experiment with them.

Note: The goal of these homework problems is to help you better understand the concept of unitary transformations and composite quantum systems, as well as to help you internalize the braket and tensor product notation. You will also revisit the Elitzur-Vaidman bomb puzzle and try to understand a different scenario related to it.

Problems:

1. (10 points) **(Ket notation)**

   (a) Let $|\phi\rangle = 3|0\rangle - 5i|1\rangle$. What number, C, should $|\phi\rangle$ be divided by to make it a "normalized" state; i.e., a unit vector? For future reference, define $|\psi\rangle = \frac{1}{C}|\phi\rangle$ to be this state vector.

   (b) Verify that $\frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle$ and $\frac{1}{\sqrt{2}}|0\rangle - \frac{i}{\sqrt{2}}|1\rangle$ form an orthonormal basis for $\mathbb{C}^2$. (These two vectors are sometimes called $|i\rangle$ and $|-i\rangle$). What are the possible outcomes and associated probabilities for measuring $|\psi\rangle$ in the $\{|i\rangle, |-i\rangle\}$ basis?

2. (10 points) **Unitary transformations.**

   Recall the definition of a unitary transformation: a linear transformation is unitary if it maps any orthonormal basis to another orthonormal basis, i.e. if it is a rotation or reflection in the complex plane.

   (a) Let $T : \mathbb{C}^n \to \mathbb{C}^n$ be a linear transformation. Show that the following three things are equivalent:

   (i) T is unitary.

   (ii) T "preserves inner products", i.e. for all $u, v \in \mathbb{C}^n$,

   $$\langle u, v\rangle = \langle Tu, Tv\rangle \ .$$

   (iii) T maps *any* unit vector (i.e. a vector with norm 1) to another vector with norm 1.

   Note that in contrast to the above, general linear transformations may change the length of a given vector or may not preserve the inner product between two given vectors.

   (b*) For a linear transformation $T : \mathbb{C}^n \to \mathbb{C}^n$, define the linear transformation $T^\dagger : \mathbb{C}^n \to \mathbb{C}^n$ to be such that, for all $u, v \in \mathbb{C}^n$,

   $$\left\langle T^\dagger u, v\right\rangle = \langle u, Tv\rangle \ .$$

   Show that $T^\dagger$ is well-defined, i.e. such a $T^\dagger$ exists and it is unique. This map $T^\dagger$ is called the adjoint of T.

   (c) Let $U : \mathbb{C}^n \to \mathbb{C}^n$ be a unitary transformation.

   (ii) Show that $U^\dagger U = UU^\dagger = I$, i.e. $U^\dagger$ is the inverse of U.

   (iii) Use what you have shown in part (a) to deduce that the matrix representation of U is such that its *columns* form an orthonormal basis.

   (iv) Use what you have shown so far in part (b) to deduce that the matrix representation of U is such that its *rows* form an orthonormal basis.

(d★) Consider the following $2^n \times 2^n$ matrix M whose rows and columns are indexed by strings $x, y \in \{0,1\}^n$ and whose entries are given by

$$M(x,y) = \prod_{i=1}^{n} (-1)^{x_i y_i}.$$

Consider the linear transformation T that the above matrix corresponds to: $T(v) = M \cdot v$. Note that T is defined over a complex vector space of 2^n dimensions where the coordinates are indexed by bitstrings $\{0,1\}^n$. Use what you have shown in part (c) above to deduce that the linear transformation that M corresponds to is a scalar multiple of a unitary. What is the scaling factor?

3. (10 points) **(Projectors and Reflections)** Let $|\psi\rangle$ and $|\phi\rangle$ be two unit vectors in $\mathbb{R}^d$. (Actually, it's fine if they're complex unit vectors, but let's keep it real.) We will be interested in $Q = |\phi\rangle\langle\psi|$, which is a $d \times d$ matrix, and can therefore be thought of as a transformation on d-dimensional vectors.

   (a) For the case d=2, explicitly work out the matrix Q in the case $|\psi\rangle = |0\rangle$ and $|\phi\rangle = |+\rangle$, and also in the opposite case $|\psi\rangle = |+\rangle$ and $|\phi\rangle = |0\rangle$.

   (b) **(Not to be submitted)** This exercise is to introduce you to the first hidden secret of the quantum club: How to perfectly hand-write the expression $|\phi\rangle\langle\psi|$. The trick is this:

   - Start by drawing the first bar and ϕ, like so: $|\phi$
   - Next — and this is the key — draw an X, like so: $|\phi\rangle\langle$
   - Finally, draw the ψ and the final bar, forming $|\phi\rangle\langle\psi|$

   If you try to draw the $\rangle$ and the $\langle$ separately, you'll never get the points to match up!

   (c) Fill in the blanks: The transformation Q maps the vector $|\psi\rangle$ to ____________ , and maps every vector orthogonal to $|\psi\rangle$ to ____________.

   (d) Suppose now that $|\psi\rangle = |\phi\rangle$. Let $P = |\psi\rangle\langle\psi|$. Describe in (geometric) words the transformation P.

   (e) Let $\mathbf{1}$ denote the identity[1] matrix in $\mathbb{R}^d$. Describe in (geometric) words the transformation $\mathbf{1} - 2P$. Your description should include the words "hyperplane perpendicular to". Prove that this transformation is unitary.

   (f) (i) Suppose we are interested in the change-of-(orthonormal-)basis operation U that takes the orthonormal basis $|\psi_1\rangle, \ldots, |\psi_d\rangle$ to the orthonormal basis $|\phi_1\rangle, \ldots, |\phi_d\rangle$. Show that U can be written as

[1] In math, it's more traditional to write I for this matrix - the diagonal matrix with 1's on the diagonal, that does nothing to a vector. But $\mathbf{1}$ is arguably a better name and is used somewhat commonly in quantum information.

$$U = |\phi_1\rangle\langle\psi_1| + \cdots + |\phi_d\rangle\langle\psi_d| \tag{1}$$

and that U is unitary.

Remark: You can verify for yourself that this gives the same answer for the matrix Q that you computed in part (a).

(ii) The identity matrix $\mathbf{1}$ is also a unitary matrix. What is the analogous decomposition (as in equation (1)) for the identity matrix?

4. (10 points) **(Quantum Anti-Zeno Effect)**

Assume you have a single qubit that you know is in the state $|0\rangle$. You really wish to change its state to $|1\rangle$. You have the ability to build any measurement device (but no unitary transformations or filters), and use it as many times as you want. How can you (with high probability) get the qubit's state changed to $|1\rangle$? (Hint: very similar to the Elitzur-Vaidman Bomb.)

Remark: This trick is called the "Quantum Anti-Zeno Effect", by analogy to an earlier discovered trick called the "Quantum Zeno Effect". In the Quantum Zeno Effect (aka Watchdog Effect), you have a qubit in state $|0\rangle$ and you wish to keep it that way; however, there is some physical process which is slowly "rotating" your qubit towards $|1\rangle$. By rapid repeated measurement in the $\{|0\rangle, |1\rangle\}$ basis, you can counteract this rotation and keep your qubit in the state $|0\rangle$ with high probability. This Quantum Zeno Effect was first described by Alan Turing!

5. (10 points) **(Composite Quantum Systems)**

Let $A \in \mathbb{R}^{n\times m}$ and $B \in \mathbb{R}^{p\times q}$. Recall that the tensor product $A \otimes B$ can be given by:

$$A \otimes B = \begin{bmatrix} A_{11}B & A_{12}B & \cdots & A_{1m}B \\ A_{11}B & A_{12}B & \cdots & A_{2m}B \\ \vdots & \vdots & \ddots & \vdots \\ A_{n1}B & A_{n2}B & \cdots & A_{nm}B \end{bmatrix}$$

is an $np \times mq$ dimensional matrix. This is also known as the Kronecker product which is a way to represent the tensor product with respect to a given basis. As is common in quantum information literature, $|01\rangle, |0\rangle \otimes |1\rangle, |0,1\rangle, |0\rangle\,|1\rangle$ all mean the same thing.

(a) Give the explicit matrix representations for the matrices $I \otimes H, H \otimes I$ and $H \otimes H$ where I is the 2×2 Identity matrix and H is the 2×2 Hadamard matrix.

(b) Compute $(I \otimes H)|0,1\rangle, (H \otimes I)|0,1\rangle$ and $(H \otimes H)|0,1\rangle$, expressing the results in the standard basis (i.e. $\sum_{i,j\in\{0,1\}} c_{ij}|i\rangle|j\rangle$ and some constants $c_{ij} \in \mathbb{C}$)

(c) Suppose $|u_1\rangle, \ldots, |u_d\rangle$ is an orthonormal basis for $\mathbb{C}^d$, and $|v_1\rangle, \ldots, |v_d\rangle$ is an orthonormal basis for $\mathbb{C}^e$. Show that the collection $|u_i\rangle \otimes |v_j\rangle$ (for all $1 \leq i \leq d, 1 \leq j \leq e$) is an orthonormal basis for $\mathbb{C}^{de}$. (Hint: exploit the braket notation to the hilt.)

(d) Let us take $n = m = d$ and $p = q = e$ and recall that the matrix $A \otimes B$ represents a linear transformation from $\mathbb{C}^{de}$ to $\mathbb{C}^{de}$. In the previous part, you have shown that the vectors $|u_i\rangle \otimes |v_j\rangle$ form an orthonormal basis. What is the matrix representation of $A \otimes B$ in this orthonormal basis? (Hint: for example, consider the matrix $|+\rangle\langle+|$: its matrix representation in the standard basis is

$$\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

but its representation in the $\{|+\rangle, |-\rangle\}$ basis is

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Express the matrix representation in terms of $A, B, |u_i\rangle, |v_j\rangle$ here.)

(e) $|\psi\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle$ is an n-qubit state where $|x\rangle = |x_1\rangle \otimes |x_2\rangle \otimes \cdots \otimes |x_n\rangle$ is a basis state for $(\mathbb{C}^2)^{\otimes n}$. Let's say we divide the state $|\psi\rangle$ into two parts as follows: we give the first j qubits to Alice and the last $n - j$ qubits to Bob. Is Alice and Bob's joint quantum state entangled? Why or why not?